

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	Semester VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory	
Academic Year	2025–2026 (Even)	Batch 2023–2027

Experiment 7

Bagging, Boosting, and Stacked Ensemble Models

Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: <https://archive.ics.uci.edu>

Ensemble Models Used

Bagging

- Base Estimator: Decision Tree
- Number of estimators
- Sampling strategy

Boosting

- AdaBoost
- Gradient Boosting

Stacked Ensemble

- Base Models: SVM, Naïve Bayes, Decision Tree
- Meta Learner: Logistic Regression

Hyperparameters to be Explored

Bagging

- `n_estimators`
- `max_samples`
- `max_features`

Boosting

- `n_estimators`
- `learning_rate`
- `max_depth`

Stacked Ensemble

- Choice of base models
- Final estimator

Graphs and Plots

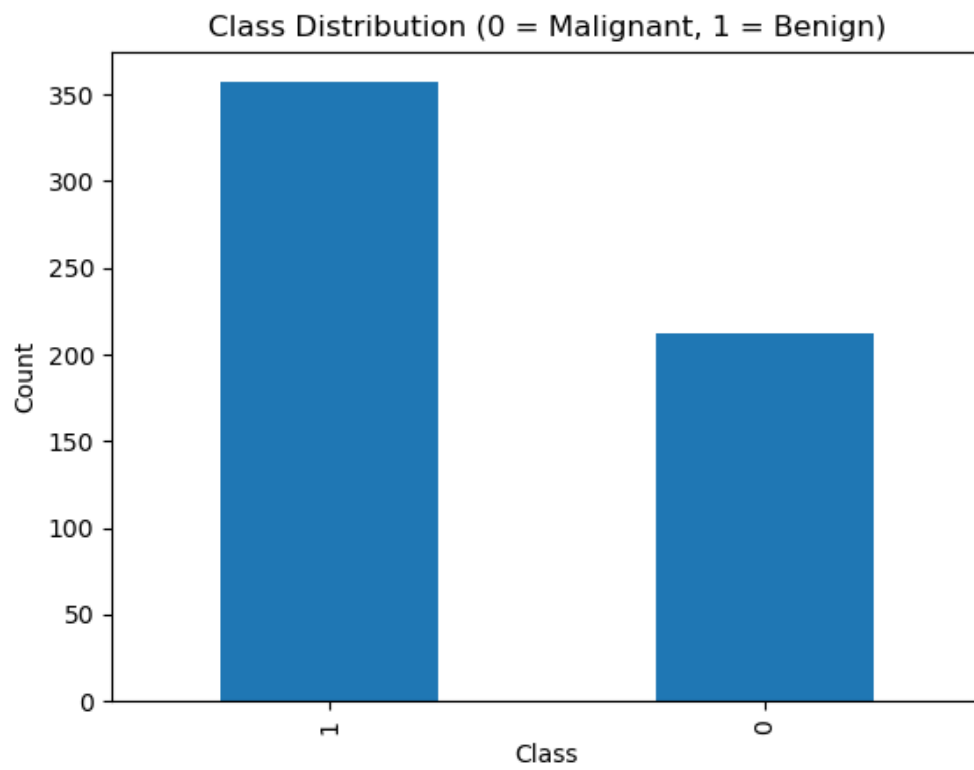


Figure 1: Class distribution

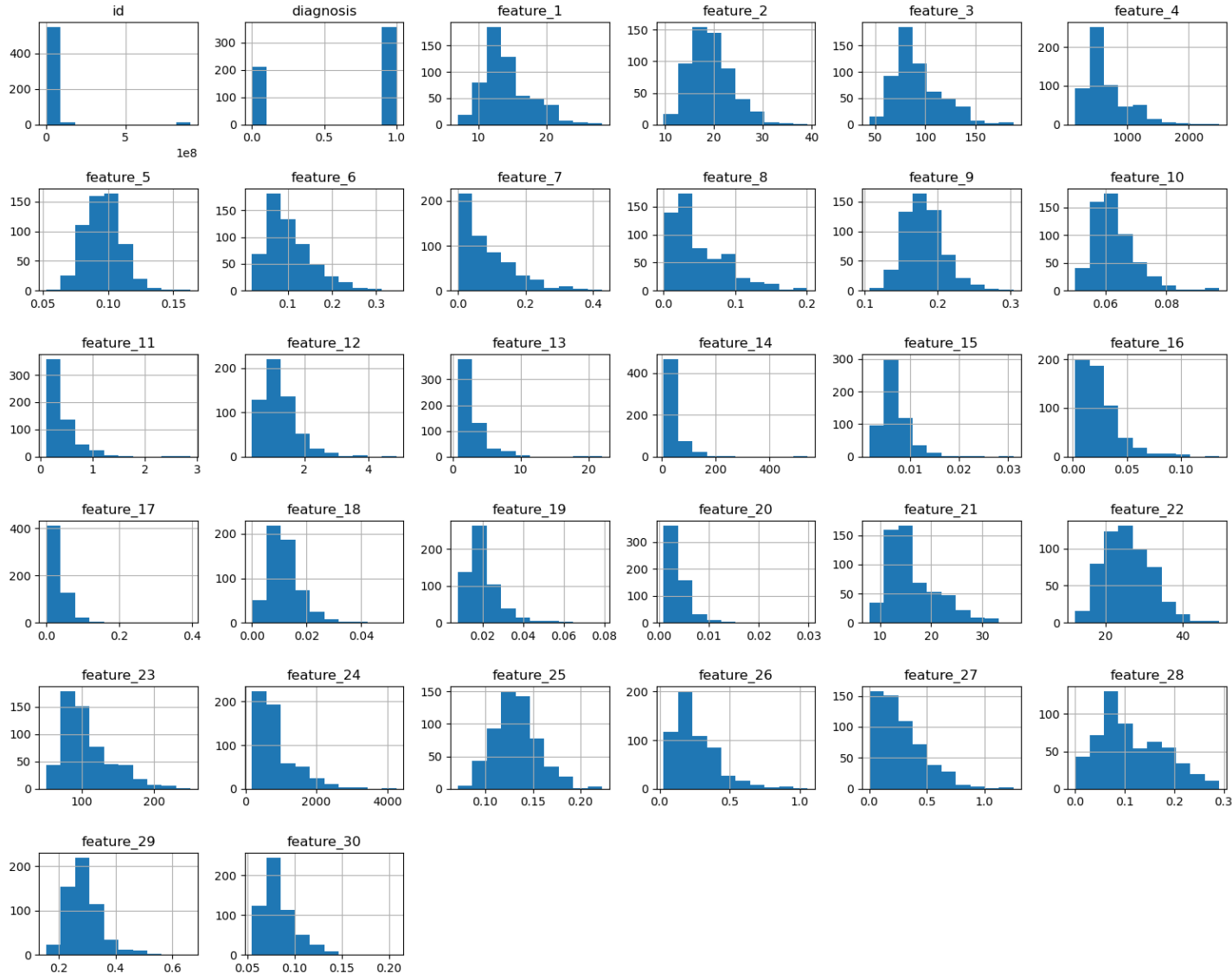


Figure 2: Histogram

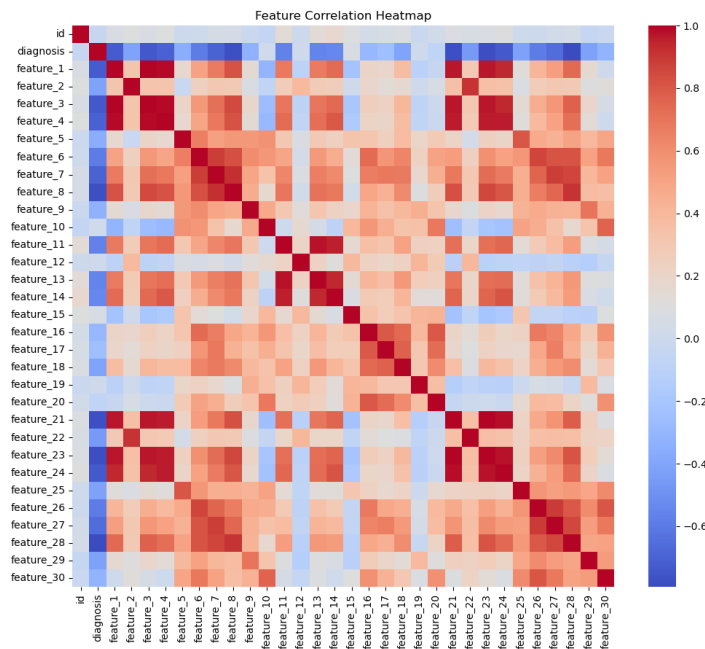


Figure 3: Feature Correlation Heatmap

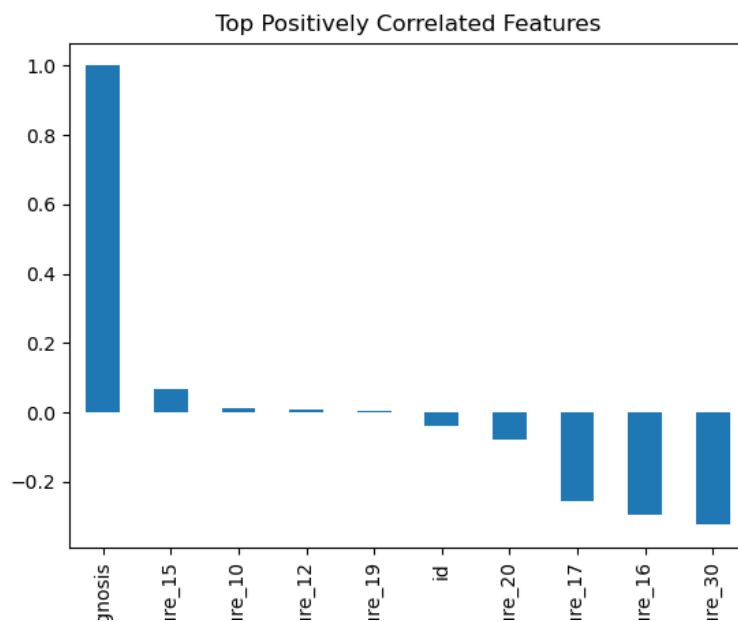


Figure 4: Top Positively Correlated Features

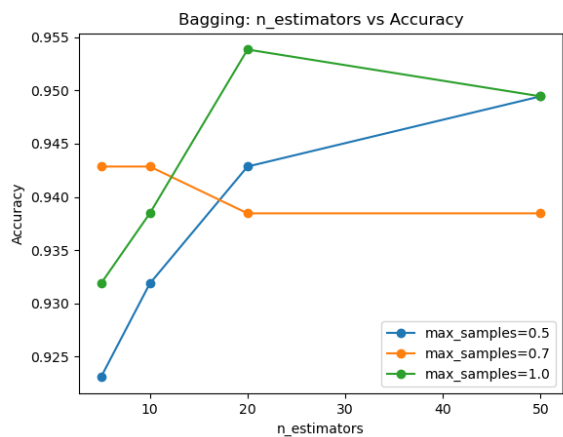


Figure 5: Bagging: $n_{estimators}$ vs Accuracy

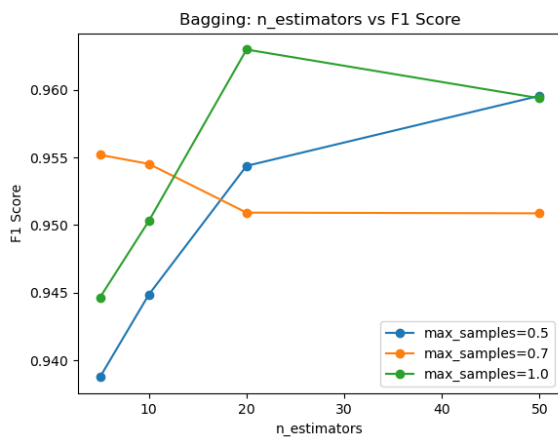


Figure 6: Bagging: $n_{estimators}$ vs F1 Score

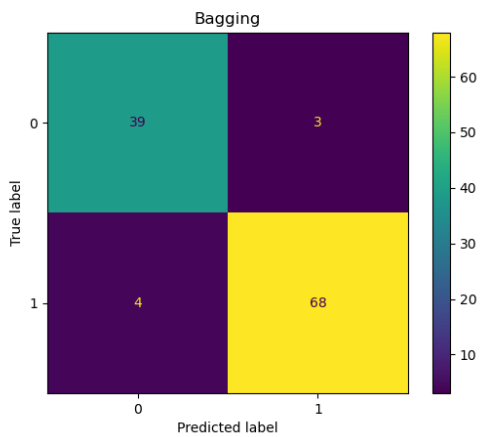


Figure 7: Bagging Confusion Matrix

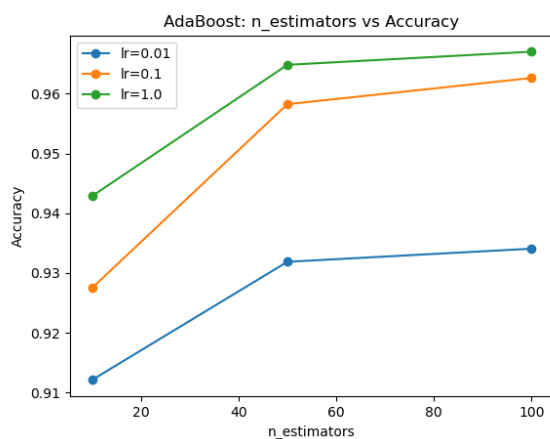


Figure 8: AdaBoost: $n_{estimators}$ vs Accuracy

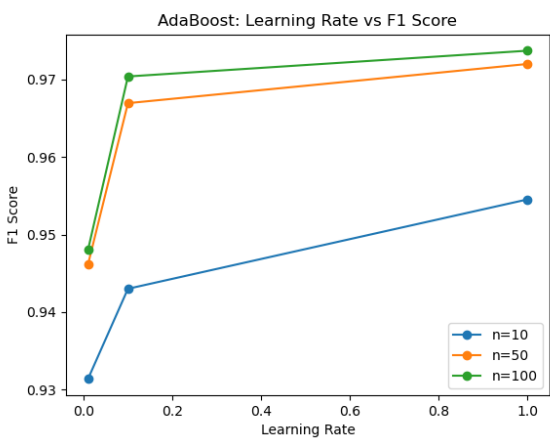


Figure 9: AdaBoost: Learning Rate vs F1 Score

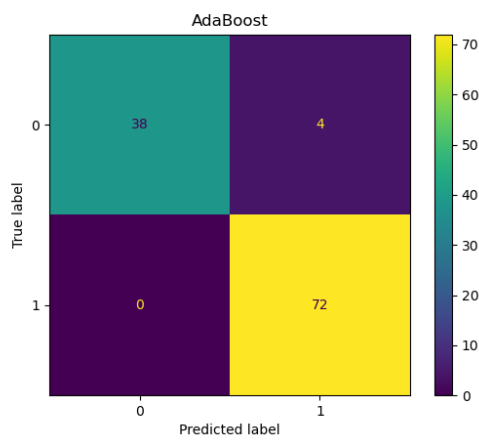


Figure 10: AdaBoost: Confusion Matrix

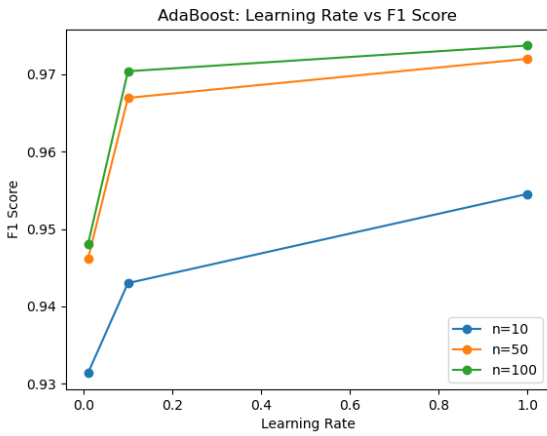


Figure 11: AdaBoost: Learning Rate vs F1 Score

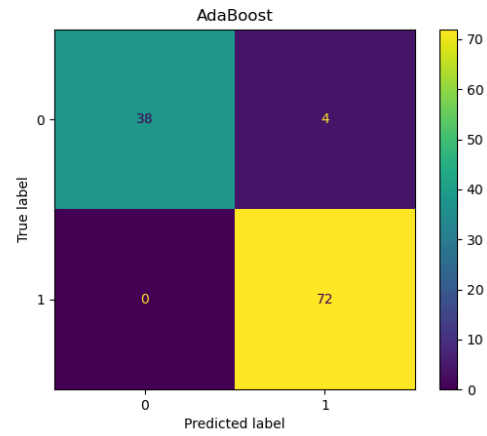


Figure 12: AdaBoost: Confusion Matrix

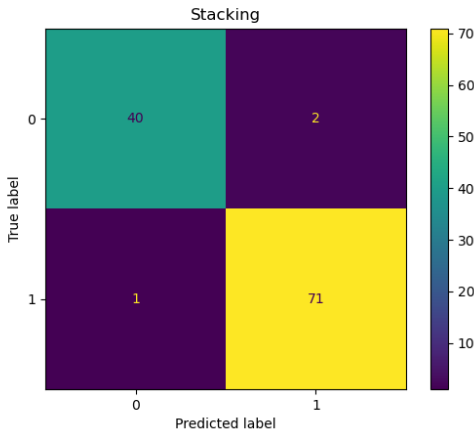


Figure 13: Stacking Ensemble: Confusion Matrix

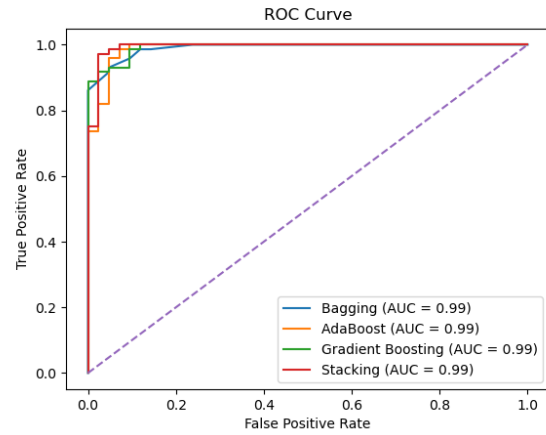


Figure 14: ROC curve

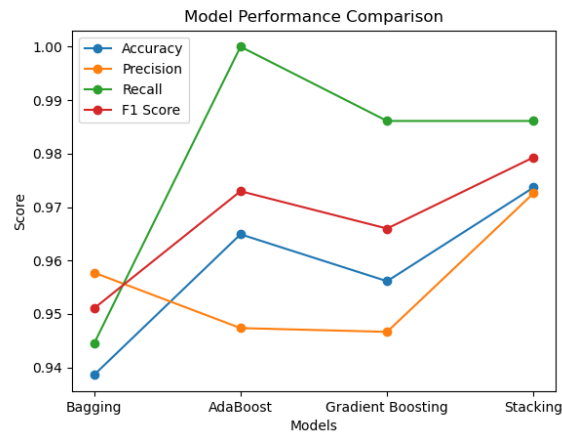


Figure 15: Model Performance Comparison

Hyperparameter Evaluation Results

Bagging Results

Table 1: Bagging Hyperparameter Evaluation

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
5	0.5	92.3077	93.8762
5	0.7	94.2857	95.5192
5	1.0	93.1868	94.4649
10	0.5	93.1868	94.4863
10	0.7	94.2857	95.4524
10	1.0	93.8462	95.0305
20	0.5	94.2857	95.4377
20	0.7	93.8462	95.0922
20	1.0	95.3846	96.2977
50	0.5	94.9451	95.9558
50	0.7	93.8462	95.0871
50	1.0	94.9451	95.9377

Boosting Results

Table 2: Boosting Hyperparameter Evaluation (Adaboost)

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.01	91.2088	93.1346
10	0.10	92.7473	94.2994
10	1.00	94.2857	95.4533
50	0.01	93.1868	94.6115
50	0.10	95.8242	96.6955
50	1.00	96.4835	97.2015
100	0.01	93.4066	94.8002
100	0.10	96.2637	97.0402
100	1.00	96.7033	97.3728

Stacked Ensemble Results

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM + Naive Bayes + Decision Tree	Logistic Regression	96.70	97.02

Performance Comparison

Table 4: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	93.8596	95.7746	94.4444	95.1049
Boosting	96.4912	94.7368	1.0000	97.2973
Stacked Ensemble	97.3684	97.2603	98.6111	97.9310

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve and AUC

Observation Questions

- How does Bagging reduce variance?
- How does Boosting address model bias?
- Why does stacking benefit from heterogeneous models?
- Which ensemble method performed best and why?

Conclusion

Bagging, Boosting, and Stacked Ensemble models were implemented and evaluated. The experiment demonstrates how different ensemble strategies improve predictive performance and generalization compared to individual models.

References

- Scikit-learn: Ensemble Methods
- Bagging Classifier
- UCI Dataset